Limits Mark Scheme A

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Award M marks for a correct method, A marks for the accurate result that follows. A correct answer with no working scores 0.

26 marks · 30 min

Q1	Q2	Q3	Q4
Q5	Q6	Q7	Q8
8	$\frac{1}{8}$	$\frac{9}{2}$	-12
$\frac{3}{4}$	2	1	$\frac{1}{2}$

1 $\lim_{x \to 4} \dfrac{x^2 - 16}{x - 4}$ $\boxed{= 8}$

$$= \lim_{x \to 4} \frac{(x-4)(x+4)}{x-4}$$ M1

Substitution gives 0/0, so factorise the difference of two squares.

$$= \lim_{x \to 4} (x+4), \ \ x \neq 4$$ A1

$$= 4 + 4 = \mathbf{8}$$ A1

2 $\lim_{x \to 0} \dfrac{\sqrt{x+16} - 4}{x}$ $\boxed{= \frac{1}{8}}$

$$= \lim_{x \to 0} \frac{(\sqrt{x+16} - 4)(\sqrt{x+16} + 4)}{x\,(\sqrt{x+16} + 4)}$$ M1

Multiply above and below by the conjugate of the numerator.

$$= \lim_{x \to 0} \frac{(x+16) - 16}{x\,(\sqrt{x+16} + 4)} = \lim_{x \to 0} \frac{1}{\sqrt{x+16} + 4}$$ A1

$$= \frac{1}{4+4} = \mathbf{\frac{1}{8}}$$ A1

3 $\lim_{x \to 3} \dfrac{x^3 - 27}{x^2 - 9}$ $= \dfrac{9}{2}$

$$= \lim_{x \to 3} \frac{(x-3)(x^2+3x+9)}{(x-3)(x+3)}$$ M1

Difference of two cubes above, difference of two squares below.

$$= \lim_{x \to 3} \frac{x^2+3x+9}{x+3}$$ A1

$$= \frac{9+9+9}{6} = \mathbf{\frac{9}{2}}$$ A1

4 $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 5x + 6}$ $= -12$

$$= \lim_{x \to 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x-3)}$$ M1

$$= \lim_{x \to 2} \frac{x^2+2x+4}{x-3}$$ A1

$$= \frac{4+4+4}{2-3} = \frac{12}{-1} = \mathbf{-12}$$ A1

WATCH FOR the denominator $2 - 3 = -1$. Answers of $+12$ lose the final A mark.

5 $\lim_{x \to \infty} \dfrac{3x^2 - 5x + 2}{4x^2 + 7x - 1}$ $= \dfrac{3}{4}$

$$= \lim_{x \to \infty} \frac{3 - 5/x + 2/x^2}{4 + 7/x - 1/x^2}$$ M1

Divide every term above and below by x^2, the highest power present.

$$\lim_{x \to \infty} \frac{1}{x} = 0 \quad \text{and} \quad \lim_{x \to \infty} \frac{1}{x^2} = 0$$ A1

$$= \frac{3-0+0}{4+0-0} = \mathbf{\frac{3}{4}}$$ A1

$\boxed{6}$ $\lim_{x\to\infty} \dfrac{\sqrt{4x^2 + 1}}{x + 3}$ $\boxed{= 2}$

$\sqrt{4x^2 + 1} \;=\; x\sqrt{4 + 1/x^2}$ for $x > 0$ M1

Take x^2 out of the root: $\sqrt{x^2} = |x| = x$ as $x \to \infty$.

$= \lim_{x\to\infty} \dfrac{\sqrt{4 + 1/x^2}}{1 + 3/x}$ A1

$= \dfrac{\sqrt{4}}{1} \;=\; \mathbf{2}$ A1

> WATCH FOR as $x \to -\infty$ the same limit is −2, because $\sqrt{x^2} = -x$ there. Worth a minute at the board.

$\boxed{7}$ $\lim_{x\to 0} \dfrac{\sqrt{1 + x} - \sqrt{1 - x}}{x}$ $\boxed{= 1}$

$= \lim_{x\to 0} \dfrac{(\sqrt{1+x} - \sqrt{1-x})(\sqrt{1+x} + \sqrt{1-x})}{x\,(\sqrt{1+x} + \sqrt{1-x})}$ M1

Multiply above and below by the conjugate.

$= \lim_{x\to 0} \dfrac{(1 + x) - (1 - x)}{x\,(\sqrt{1+x} + \sqrt{1-x})} \;=\; \lim_{x\to 0} \dfrac{2x}{x\,(\sqrt{1+x} + \sqrt{1-x})}$ A1

$= \lim_{x\to 0} \dfrac{2}{\sqrt{1+x} + \sqrt{1-x}}$ (cancel $x \neq 0$) A1

$= \dfrac{2}{1 + 1} \;=\; \mathbf{1}$ A1

$\boxed{8}$ $\lim_{x\to\infty} \left(\sqrt{x^2 + x} - x \right)$

$\boxed{= \dfrac{1}{2}}$

$$= \lim_{x\to\infty} \frac{(\sqrt{x^2+x} - x)(\sqrt{x^2+x} + x)}{\sqrt{x^2+x} + x}$$ M1

The form is ∞ − ∞, so rationalise: multiply and divide by the conjugate.

$$= \lim_{x\to\infty} \frac{(x^2 + x) - x^2}{\sqrt{x^2+x} + x} = \lim_{x\to\infty} \frac{x}{\sqrt{x^2+x} + x}$$ A1

$$= \lim_{x\to\infty} \frac{1}{\sqrt{1 + 1/x} + 1} \quad \text{(divide by } x > 0\text{)}$$ A1

$$= \frac{1}{1 + 1} = \mathbf{\frac{1}{2}}$$ A1

WATCH FOR students who write ∞ − ∞ = 0. State the indeterminate form before rationalising.